



Free Cybersecurity Learning Resources

Five hands-on platforms for building practical cybersecurity skills

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Resources at a Glance

A mix of web security, blue team, CTF, and exploitation practice

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|---|--------------------------|---------------------------------|
| 1 | PortSwigger | Web application security |
| 2 | CyberDefenders | Blue team & incident response |
| 3 | CyLab | CTF-style security challenges |
| 4 | pwn.college | Systems & exploitation |
| 5 | Exploit Education | Low-level software exploitation |

PortSwigger Web Security Academy

Free interactive web application security labs

PortSwigger Web Security Academy teaches web application security through lessons and hands-on labs.

Skills Practiced

- SQL injection
- Cross-site scripting (XSS)
- Authentication
- Access control

<https://portswigger.net/web-security/getting-started>

CyberDefenders

Hands-on defensive security and blue-team training: Not completely free but hundreds of free labs.

CyberDefenders provides practical blue-team challenges based on realistic security investigations. Learners analyze evidence, investigate suspicious activity, and develop skills commonly used by SOC analysts and incident responders.

Skills Practiced

- Digital forensics
- Threat hunting
- Malware analysis
- Incident response

<https://cyberdefenders.org/dashboard/>

CyLab Security Academy

CTF-style cybersecurity education and practice

CyLab Security Academy uses cybersecurity challenges to help learners build technical skills through practice. Its challenge-based format gives students experience solving security problems rather than only reading about them.

Skills Practiced

- Cryptography
- Forensics
- Web exploitation
- Binary exploitation

<https://learn.cylabacademy.org/dashboard>

pwn.college

Free hands-on systems security education

pwn.college teaches cybersecurity through interactive challenges and structured modules. Learners work in practical environments to develop Linux, system security, reverse engineering, and exploitation skills.

Skills Practiced

- Linux & command line
- System security
- Reverse engineering
- Binary exploitation

<https://pwn.college/welcome/welcome/>

Exploit Education — Phoenix

Vulnerable environments for learning software exploitation

Phoenix is a training environment designed to teach the fundamentals of software exploitation. Its exercises help learners understand how memory-related vulnerabilities occur and how attackers can take advantage of insecure programs.

Skills Practiced

- Buffer overflows
- Memory corruption
- Format strings
- Low-level debugging

<https://exploit.education/phoenix/>